Design Definition:

Mountain Hub
Electric Mobility System

Group 8

- Anna Baranova
- Giovanni Sciarra
- Olaf Grzegory
- Naren Muthuraj
- Sergio Alonso Marquez Guillen
- Gabriela Alvarez Jaramillo

Professors :
Gatti Riccardo
Longana Marco Luigi
Schreiber Daniel Bruno Josef
Rozzoni Marcio



BRIEF & DESING REQUIREMENTS

‘An electric-assisted cargo vehicle to make daily duty mountain **resupply logistics** feel **effortless** and **weightless**. designed for **high-gradient, harsh terrain** to support **manual labour** and **reduce noise and air pollution** in remote mountain environments. **‘**On-site' machine facilitates the **transport of diverse materials** including **construction tools, timber, solar panels and food supplies.**

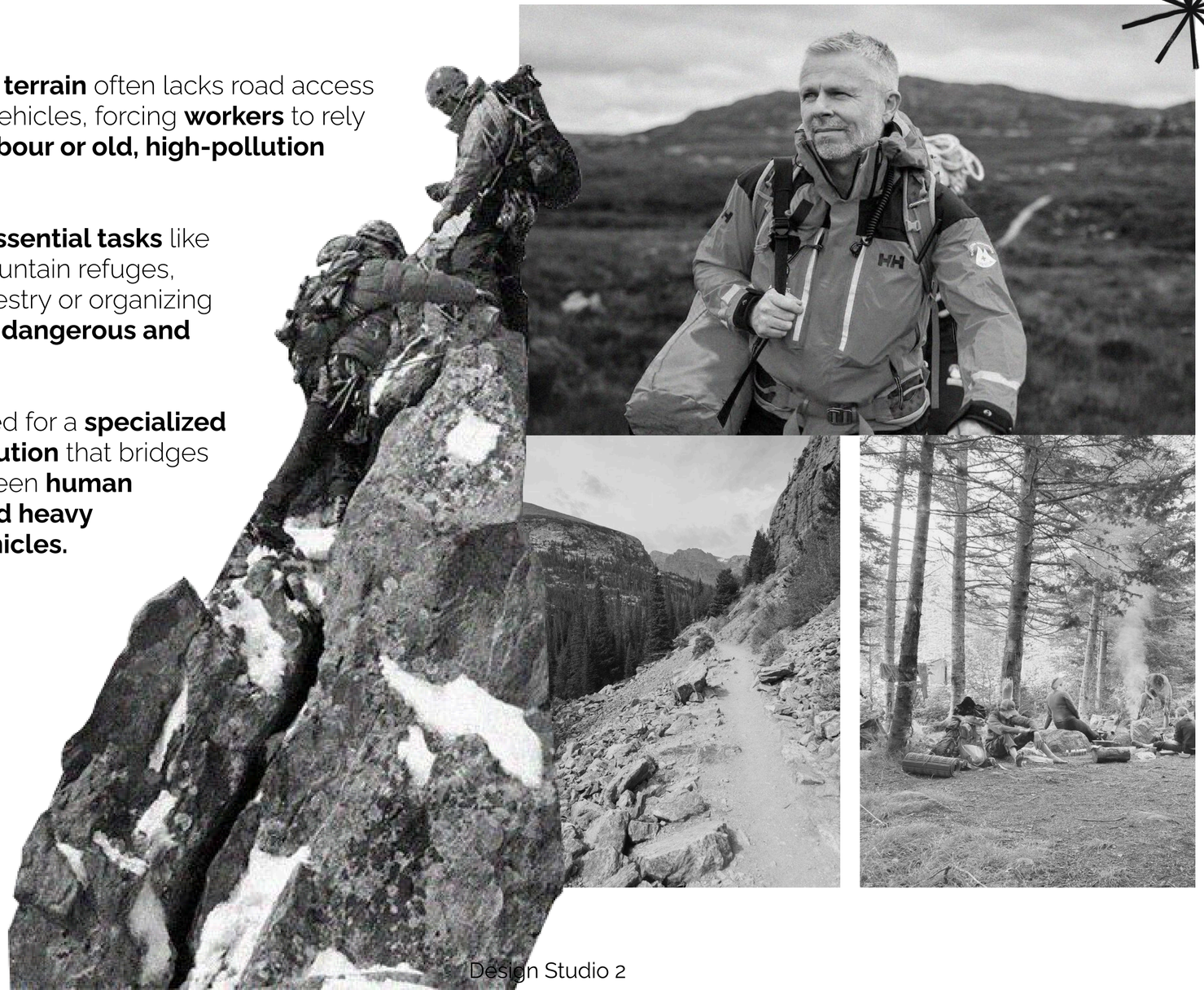
USER ANALYSIS

WHY

Mountainous terrain often lacks road access for standard vehicles, forcing **workers** to rely on **manual labour or old, high-pollution machinery**.

This **makes essential tasks** like supplying mountain refuges, managing forestry or organizing sports events **dangerous and inefficient**.

There is a need for a **specialized transport solution** that bridges the gap between **human portaging and heavy industrial vehicles.**



WHO

1. The Mountain Refuge Worker
The Mission: Keeping the system running. They deal with high-volume, repetitive
supply runs (food, gas canisters, fresh linens).

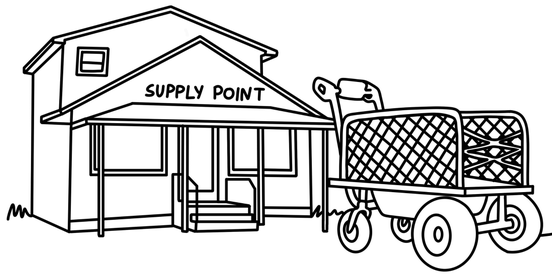
Main Tasks:

- Restocking supplies
- Cleaning Main Tasks:
- Waste management
- Infrastructure maintenance (plumbing, electric works)
- Emergency support

Average radius of work:
Maximum of 500m around the refuge with Average radius of work exceptions for occasional delivery of food supplies to a neighbouring hut or in order to provide first aid for tourists.

STORYBOARD

Source of departure



Inspection of the overall state of the vehicle (ex. wheels and battery status)

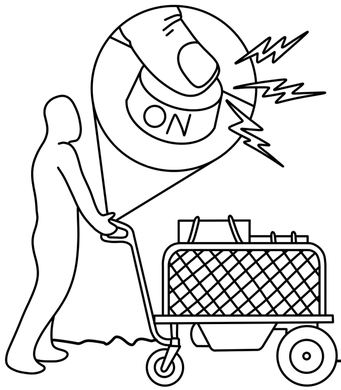


Load cargo



Secure cargo





In order to begin Journey
Turn ON vehicle

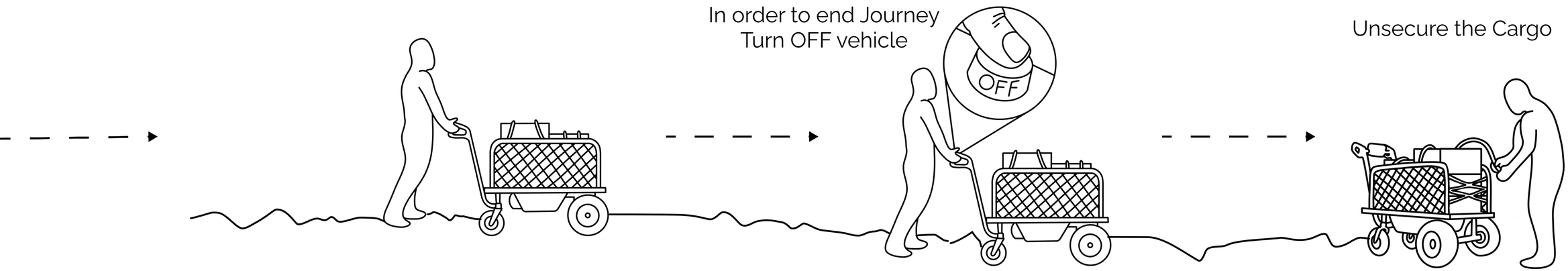
Throttle the cart



Drive the cart

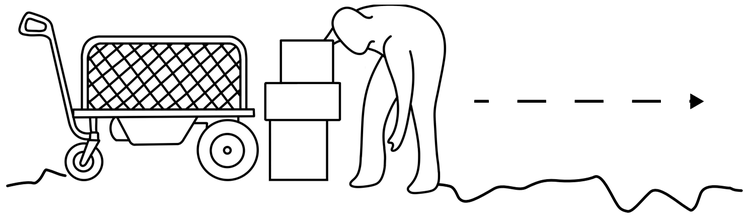


STORYBOARD

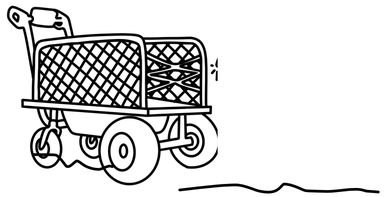


In order to end Journey
Turn OFF vehicle

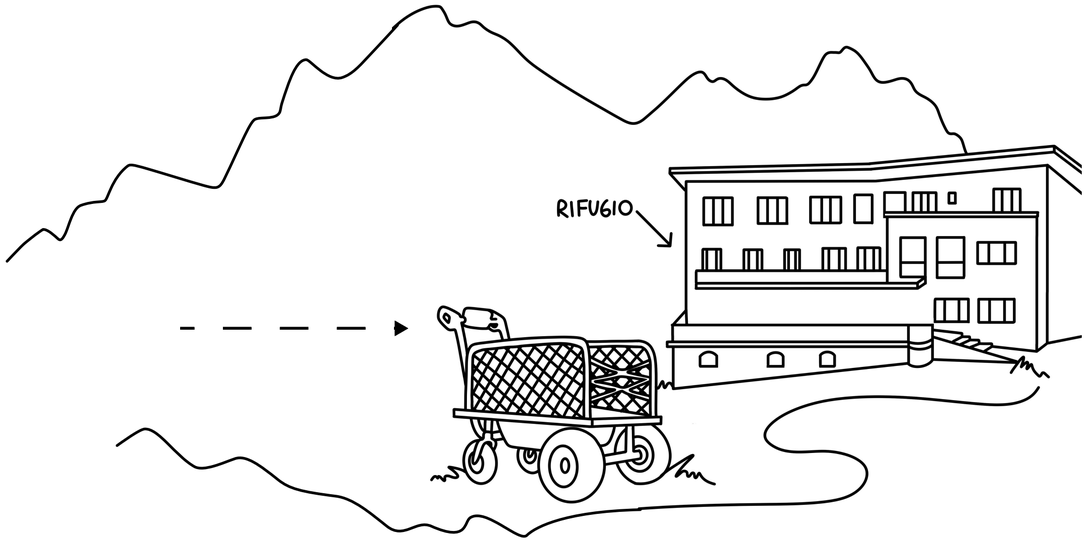
Unsecure the Cargo



Unload cargo



Recharge the cart



Secure vehicle at point of arrival

CRITICAL POINTS

1. **Terrain Capability:**
Ability to drive in various terrains where conventional vehicles cannot access. Driving securely, on a steep incline

2. **Weatherproof:**
Waterproof and must operate in both low and high temperatures

3. **Stability:**
Good suspension. Should not be possible to tip over

4. **Portability:**
Light enough to be pushed over rocks by 1/2 people. Possibly, easy disassembly for better transportation

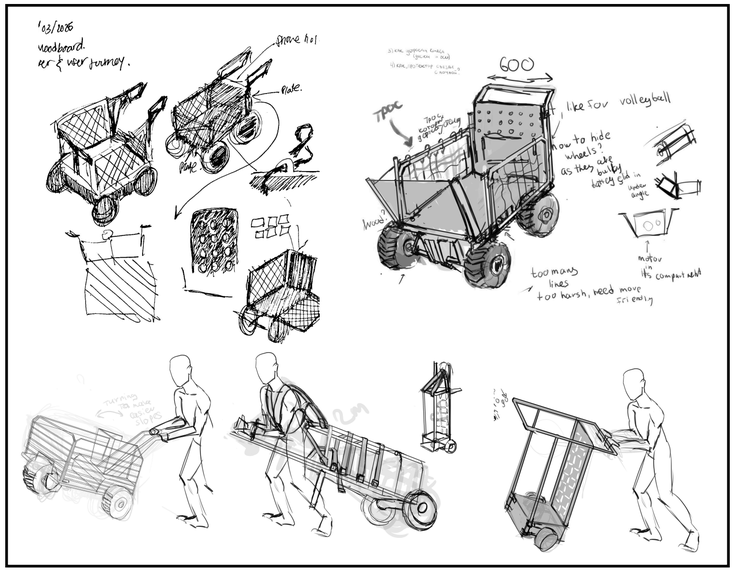
5. **Cargo Capacity:**
Max. load 200kg
Adjustable table for cargo (small and large). Proper mechanisms to secure the cargo

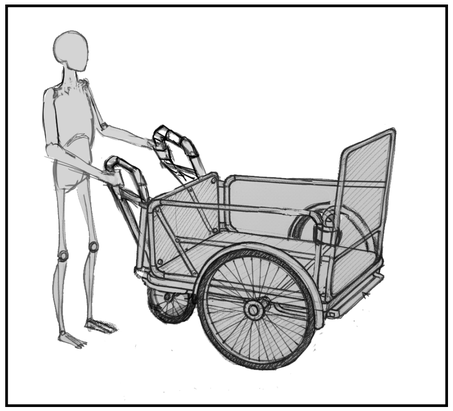
6. **Strength & Durability:**
Resistance to permanent deformation and easy access for maintenance

7. **Battery:**
Capacity good enough to operate under harsh terrain and weather conditions

8. **Ergonomics:**
Adjustable steering handle and button-operated user interface

PROGRESS TIMETABLE

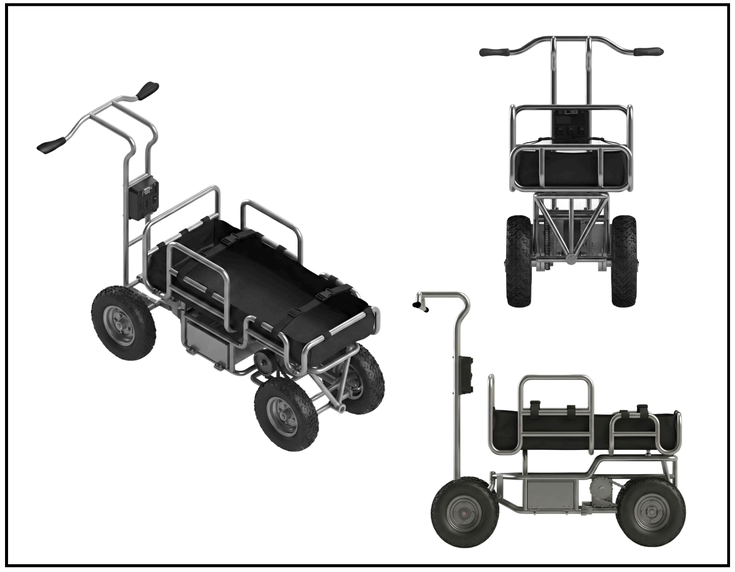














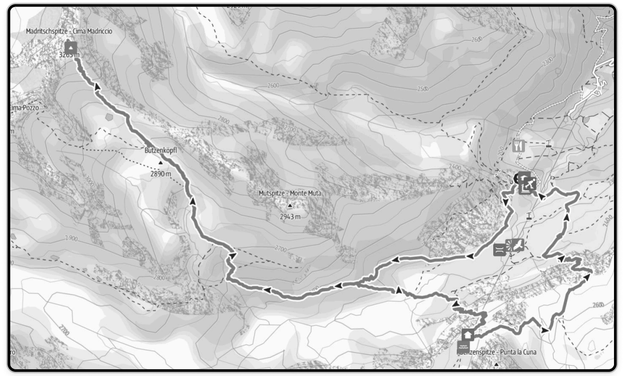


FINAL DESIGN

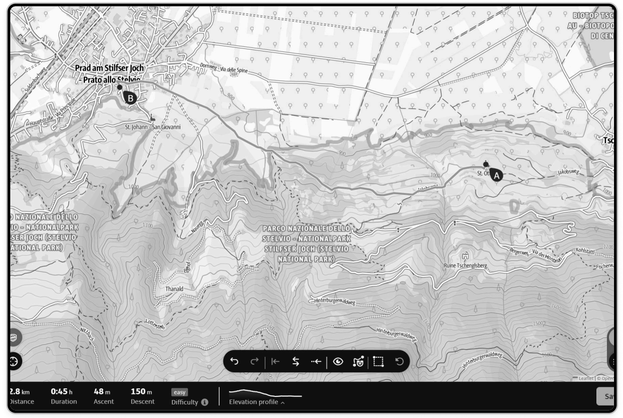


MAIN ISSUES AND SOLUTIONS

User & Environment specifications









What is the exact use of the cart?
Distance to cover daily?
Goods to transport?
Road difficulty and slope?
Who is the exact user of the cart?

fédération française des clubs alpins et de montagne

Club Arc Alpin

MAIN ISSUES AND SOLUTIONS

Engineering calculations

VARIABLE INPUTS

MASS
ACCEL
SLOPE
TERRAIN

STAGE 1
MECHANICAL

Forces, torque, gear ratio
Core of the engine

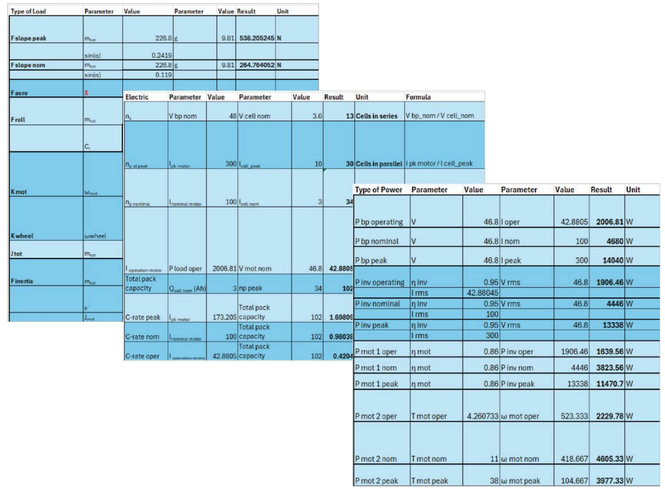
STAGE A1
BATTERY

No. cells, runtime

STAGE A2
POWER

Power demand and supply

RESULTS



TERRAIN
TIRE TYPE
TIRE PRESSURE
SLOPE

STAGE B1
TIRE AND TERRAIN DATA EXTRACTION

Specification of all the parameters

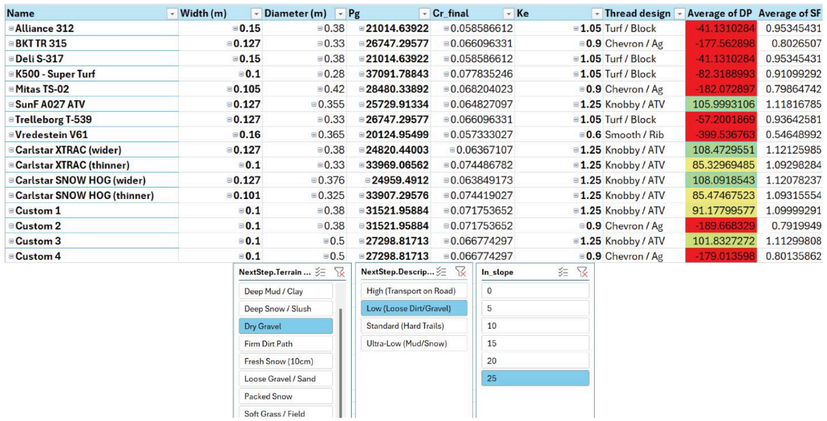
STAGE B2
1 OPTION ENGINE

Manual input engine only
for 1 tire type

STAGE B3
MULTIPLE OPTION ENGINE

Automatic input engine for
multiple types of tires

RESULTS



Name	Width (m)	Diameter (m)	Pg	Cr_final	Ke	Thread design	Average of DP	Average of SF
Alliance 312	0.15	0.38	21014.63922	0.058586612	1.05	Turf / Block	-41.1310284	0.95345431
BKT TR 315	0.127	0.33	26747.29577	0.066096331	0.9	Chevron / Ag	-177.562898	0.8026507
Deli S-317	0.15	0.38	21014.63922	0.058586612	1.05	Turf / Block	-41.1310284	0.95345431
K500 - Super Turf	0.1	0.28	37091.78843	0.077835246	1.05	Turf / Block	-82.3188993	0.91099292
Mitas TS-02	0.105	0.42	28480.33892	0.068204023	0.9	Chevron / Ag	-182.072897	0.79864742
SunF A027 ATV	0.127	0.355	25729.91334	0.064827097	1.25	Knobby / ATV	105.9993106	1.11816785
Trelleborg T-539	0.127	0.33	26747.29577	0.066096331	1.05	Turf / Block	-57.2001869	0.93642581
Vredestein V61	0.16	0.365	20124.95499	0.057333027	0.6	Smooth / Rib	-399.536763	0.54648992
Carlstar XTRAC (wider)	0.127	0.38	24820.44003	0.06367107	1.25	Knobby / ATV	108.4729551	1.12125985
Carlstar XTRAC (thinner)	0.1	0.33	33969.06562	0.074486782	1.25	Knobby / ATV	85.32969485	1.09298284
Carlstar SNOW HOG (wider)	0.127	0.376	24959.4912	0.063849173	1.25	Knobby / ATV	108.0918543	1.12078237
Carlstar SNOW HOG (thinner)	0.101	0.325	33907.29576	0.074419027	1.25	Knobby / ATV	85.47467523	1.09315554
Custom 1	0.1	0.38	31521.95884	0.071753652	1.25	Knobby / ATV	91.17799577	1.09999291
Custom 2	0.1	0.38	31521.95884	0.071753652	0.9	Chevron / Ag	-189.668329	0.7919949
Custom 3	0.1	0.5	27298.81713	0.066774297	1.25	Knobby / ATV	101.8327272	1.11299808
Custom 4	0.1	0.5	27298.81713	0.066774297	0.9	Chevron / Ag	-179.013598	0.80135862

MAIN ISSUES AND SOLUTIONS

Product Optimization for Manufacturing

Individual detailed studies

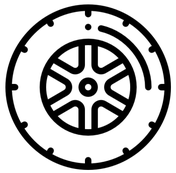
Suspension

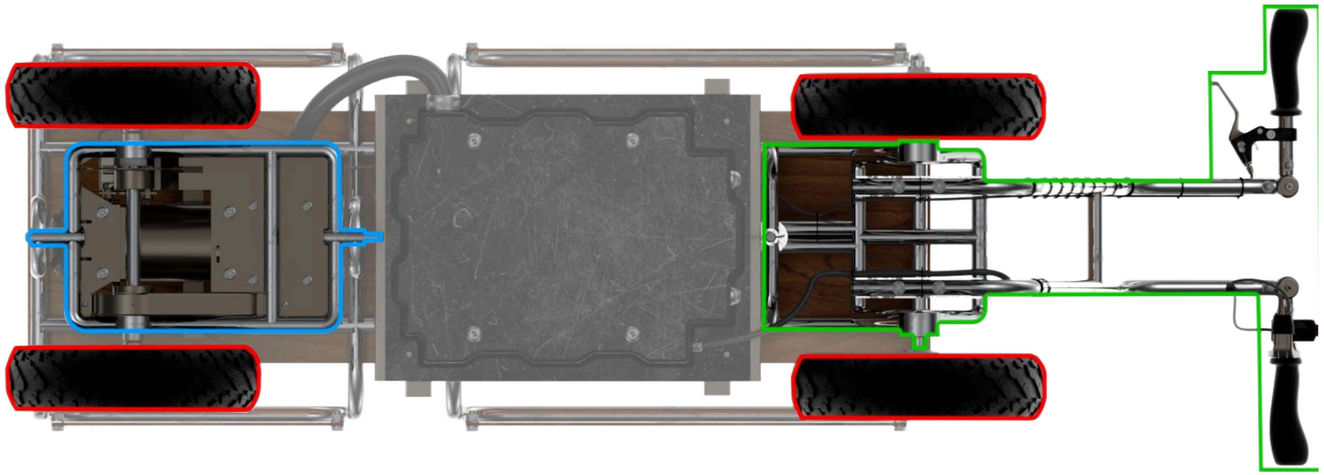


Steering

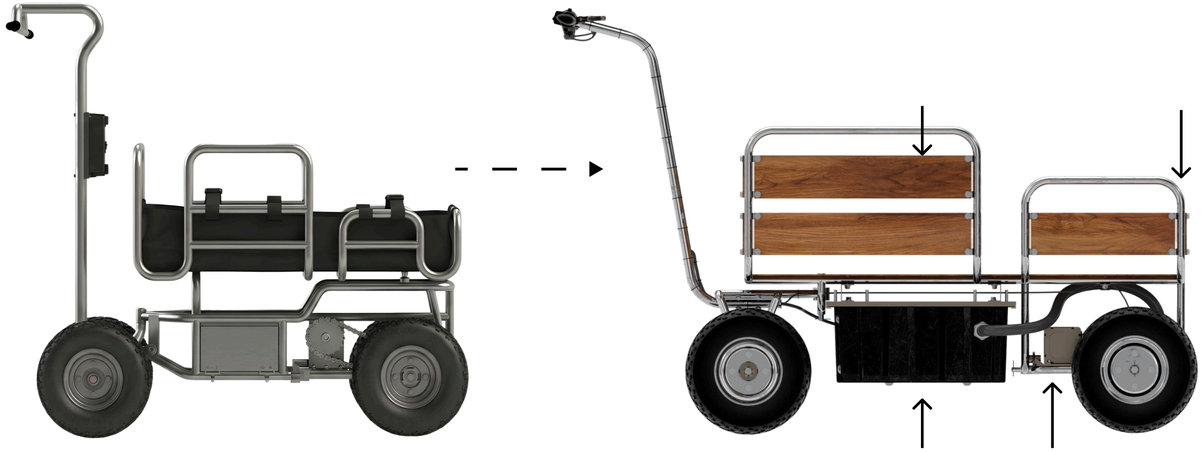


Traction





Unification of individual components



MAIN ISSUES AND SOLUTIONS

BOM Classification

Difficulties faced

- Determining which parts should be manufactured, purchased, or assembled.
- Defining assembly hierarchy and **parent-child relationships**.
- Identifying overlooked components such as fasteners, bushes, bearings, spacers, and mounting hardware.
- Establishing a logical part numbering system.
- Maintaining consistency across multiple subsystems designed by different team members.
- Converting design intent into information that could realistically be used by manufacturing and procurement teams.

Our approach to it

- Conducted repeated assembly-level reviews and BOM cross-checks with CAD
- Discussed and resolved differences in component structuring within the team
- Developed a **standardized system** for:
  - Part naming and numbering
  - Assembly hierarchy definition
  - Component classification
- Used faculty feedback to align with standard engineering documentation practices


THANK YOU!

MAIN ISSUES AND SOLUTIONS

Engineering calculations

STAGE 1
MECHANICAL

Variables:
Mass
Accel
Slope
Terrain

Forces, torque, gear ratio, core of the engine

STAGE A1
BATTERY

A1

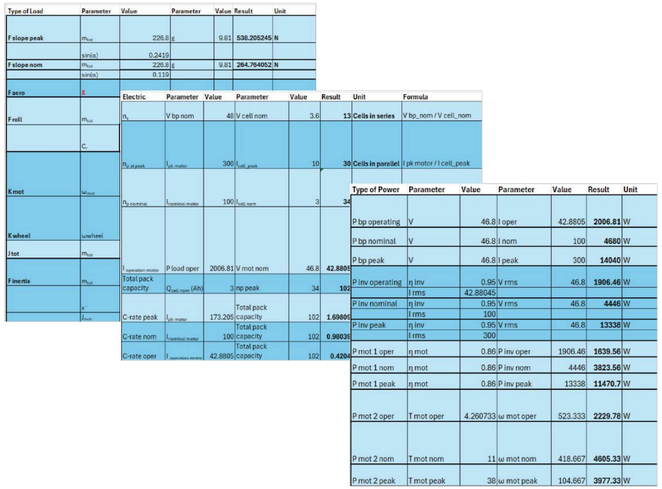
No cells. runtime

STAGE A2
BATTERY

A2

Power demand and suply

RESULTS



STAGE B1
TA AND TERRAIN DATA EXTRACTION

Terrain
Tire type
Tire pressure
Slope

B1

Specification and terrain data extraction

STAGE B2
1 OPTION ENGINE

A1

Manual input engine onlu for 1 tire type

STAGE B3
MULTIPLE OPTION ENGINE

A2

Automatic input engine for multiple types of tires

RESULTS

Name	Width (m)	Diameter (m)	Pg	Cr_final	Ke	Thread design	Average of DP	Average of SF
Alliance 312	0.15	0.38	21014.63922	0.058586612	1.05	Turf / Block	-41.1310284	0.95345431
BKT TR 315	0.127	0.33	26747.29577	0.066096331	0.9	Chevron / Ag	-177.562898	0.8026507
Deli S-317	0.15	0.38	21014.63922	0.058586612	1.05	Turf / Block	-41.1310284	0.95345431
K500 - Super Turf	0.1	0.28	37091.78843	0.077835246	1.05	Turf / Block	-82.3188993	0.91099292
Mitas TS-02	0.105	0.42	28480.33892	0.068204023	0.9	Chevron / Ag	-182.072897	0.79864742
SunF A027 ATV	0.127	0.355	25729.91334	0.064827097	1.25	Knobby / ATV	105.9993106	1.11816785
Trelleborg T-539	0.127	0.33	26747.29577	0.066096331	1.05	Turf / Block	-57.2001869	0.93642581
Vredestein V61	0.16	0.365	20124.95499	0.057333027	0.6	Smooth / Rib	-399.536763	0.54648992
Carlstar XTRAC (wider)	0.127	0.38	24820.44003	0.06367107	1.25	Knobby / ATV	108.4729551	1.12125985
Carlstar XTRAC (thinner)	0.1	0.33	33969.06562	0.074486782	1.25	Knobby / ATV	85.32969485	1.09298284
Carlstar SNOW HOG (wider)	0.127	0.376	24959.4912	0.063849173	1.25	Knobby / ATV	108.0918543	1.12078237
Carlstar SNOW HOG (thinner)	0.101	0.325	33907.29576	0.074419027	1.25	Knobby / ATV	85.47467523	1.09315554
Custom 1	0.1	0.38	31521.95884	0.071753652	1.25	Knobby / ATV	91.17799577	1.09999291
Custom 2	0.1	0.38	31521.95884	0.071753652	0.9	Chevron / Ag	-189.668379	0.7919949
Custom 3	0.1	0.5	27298.81713	0.066774297	1.25	Knobby / ATV	101.8327272	1.11299808
Custom 4	0.1	0.5	27298.81713	0.066774297	0.9	Chevron / Ag	-179.013598	0.80135862

